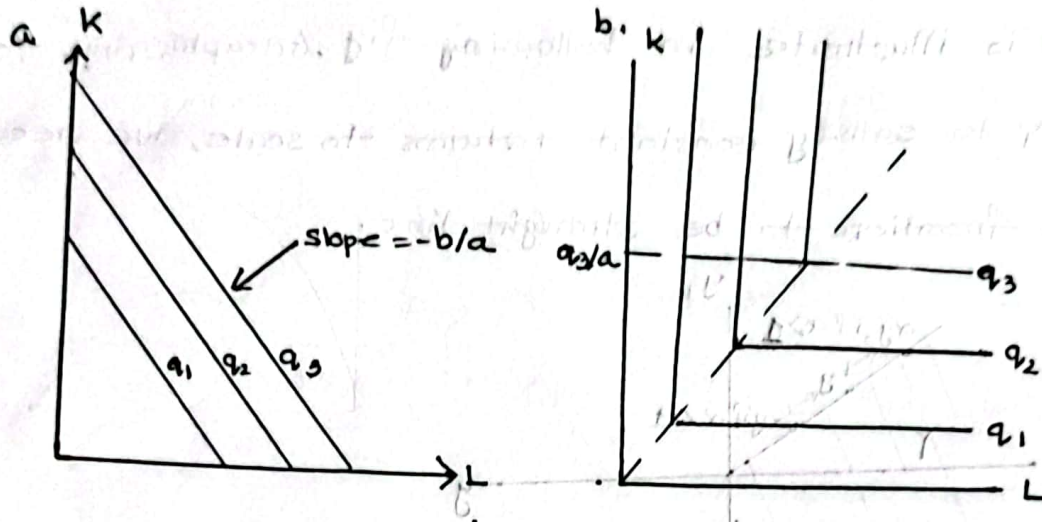


Panel a of the following fig illustrates a linear production function in which labor and capital are perfect substitutes in production process. It implies that $MRTS_{L,K}$ is constant in k/L and hence the elasticity of substitution yields an infinitely large number

$$\sigma = \infty$$



Let us next examine the other polar case in which both inputs must be used in fixed proportion, as $q = \min\{ak, bL\}$, where $a, b > 0$.

In this case the $MRTS_{L,K}$ changes dramatically from infinite (for labor amounts below the kink in the isoquants as depicted in panel b) to zero. This implies the change in $MRTS_{L,K}$ is infinite, yielding as a consequence a zero elasticity of substitution, namely $\sigma = 0$.

21. Define profit function. If the production set Y is closed and have fixed disposal, then what ^{properties} responsibilities do the profit function satisfy?

If we consider a profit maximization problem, $\max p \cdot y$ subject to $y \in Y$, the value function resulting from this profit maximization problem, $\pi(p)$, denoted

ted as the "profit function" of the firm, describes for every price p the highest amount of profits, chosen by the profit-maximizing production plan $y(p)$. More formally, we can define profit function as:

$$\pi(p) = p \cdot y(p);$$

that is the firm's profit evaluated at $y = y(p)$.

Now, let us describe some properties of the profit function $\pi(p)$.

Let us assume that the production set Y is closed and satisfies the free disposal property:

1. Homogeneity: The profit function $\pi(p)$ is homogeneous of degree one in prices. That is increasing the prices of all inputs and outputs by the same factor λ produces a proportional increase in the firm's profits, $\pi(\lambda p) = \lambda \pi(p)$. The profit function can be expressed as $\pi(p) = p \cdot q - w \cdot z$, where all $\perp$ inputs (in vector z) and the output q are evaluated at their profit-maximizing amounts. Scaling all prices by common factor λ , we obtain

$$\lambda p \cdot q - \lambda w \cdot z = \lambda (p \cdot q - w \cdot z) = \lambda \cdot \pi(p)$$

which shows that the firm's profit increases in the same proportion (λ) as input and output prices, thus implying that homogeneity of degree one.

2. Convex in price: The profit function $\pi(p)$ is convex. The following fig describes this property. For a price vector p , the profit-maximizing production plan is $y(p)$, yielding an associated profit function $\pi(p)$. Similarly, for a price vector p' , the profit maximizing production plan is $y(p')$, yielding an associated profit function $\pi(p')$.

3. Alternative representation: If a production set Y is convex, then

$$Y = \{y \in \mathbb{R}^L : py \leq \pi(p) \text{ for all } p \gg 0\}$$

Intuitively, the production set Y can be alternatively represented by the "dual" set. It specifies that, for any given price vector p , all production vectors y generate less profit py than the profit maximizing production plan $y(p)$, which entails a profit function $\pi(p) = py(p)$. A graphical representation of this property is shown in following fig:

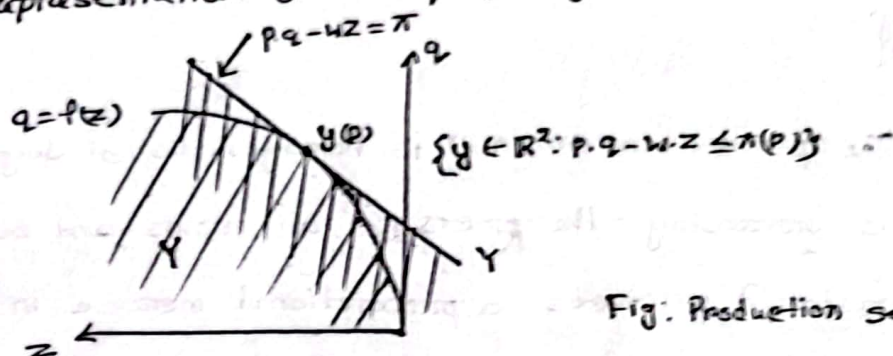


Fig: Production set Y is convex.

As the fig shows, in a convex production set Y , the supply correspondence $y(p)$ maximizes profits and the associated isoprofit line is $\pi = p q - w z$. Since all production plans (z, q) below this isoprofit line is ~~isoprofit~~ yield a lower profit level for the firm $p q - w z \leq \pi(p)$. Alternatively, the isoprofit line $\pi = p q - w z$ can be represented by

$$q = \frac{\pi}{p} - \frac{w}{p} z$$

If the price vector w/p is constant, meaning input (output) prices are unaffected by different level of input usage, then the slope of the isoprofit line is constant in z . Therefore the set of (z, q) pairs below profit function $\pi(p)$ define a convex set.

22. Consider a consumer with the utility function $u(x_1, x_2) = x_1^{1/4} x_2^{1/2}$

i) Derive the ordinary and compensated demand function.

ii) Consider $\{p^0, m^0\} = \{4, 4, 3000\}$ a) compute the indirect utilities and
 $\{p^1, m^0\} = \{2, 4, 300\}$ cost

iii) Calculate EV, EV (ges) and examine their relationship and comment on the nature of the commodities.

i) Given, Cobb-Douglas utility function, $u(x_1, x_2) = x_1^{1/4} \cdot x_2^{1/2}$
Budget constraint, $m = p_1 x_1 + p_2 x_2$

Waltzian / Marshallian demand function → Maximize utility

$$x_1(p, w) = \frac{\alpha w}{(\alpha + \beta) p_1} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{2}} \cdot \frac{m}{p_1} = \frac{\frac{1}{4}}{\frac{3}{4}} \cdot \frac{m}{p_1} = \frac{1}{3} \cdot \frac{m}{p_1}$$

$$x_2(p, w) = \frac{\beta w}{(\alpha + \beta) p_2} = \frac{\frac{1}{2}}{\frac{1}{4} + \frac{1}{2}} \cdot \frac{m}{p_2} = \frac{\frac{1}{2}}{\frac{3}{4}} \cdot \frac{m}{p_2} = \frac{2}{3} \cdot \frac{m}{p_2}$$

Compensated / Hicksian demand function → Minimize expenditure

$$h_1(p, u) = \left(\frac{\alpha}{\beta} \cdot \frac{p_2}{p_1} \right)^{\frac{\beta}{\alpha + \beta}} \cdot u^{\frac{1}{\alpha + \beta}} = \left(\frac{\frac{1}{4}}{\frac{1}{2}} \cdot \frac{p_2}{p_1} \right)^{\frac{1/2}{1/4 + 1/2}} \cdot u^{\frac{1}{1/4 + 1/2}} = \left(\frac{p_2}{2p_1} \right)^{\frac{2}{3}} \cdot u^{\frac{4}{3}}$$

$$h_2(p, u) = \left(\frac{\beta}{\alpha} \cdot \frac{p_1}{p_2} \right)^{\frac{\alpha}{\alpha + \beta}} \cdot u^{\frac{1}{\alpha + \beta}} = \left(\frac{1/2}{1/4} \cdot \frac{p_1}{p_2} \right)^{\frac{1/4}{1/4 + 1/2}} \cdot u^{\frac{1}{1/4 + 1/2}} = \left(\frac{2p_1}{p_2} \right)^{\frac{1}{3}} \cdot u^{\frac{4}{3}}$$

ii) Indirect Utility:

$$\text{Indirect utility } V = v(p, w) = \left(\frac{\alpha}{\alpha + \beta} \cdot \frac{w}{p_1} \right)^{\alpha} \left(\frac{\beta}{\alpha + \beta} \cdot \frac{w}{p_2} \right)^{\beta} = \left(\frac{\alpha}{p_1} \right)^{\alpha} \left(\frac{\beta}{p_2} \right)^{\beta} \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta}$$

$$\text{Indirect utility at initial prices } u^* = \left(\frac{1/4}{1} \right)^{1/4} \left(\frac{1/2}{1} \right)^{1/2} \left(\frac{300}{1/4 + 1/2} \right)^{3/4 + 1/2} = 15.81$$

$$\text{Indirect } \gg \gg \text{ final } \gg u^1 = \left(\frac{1/4}{2} \right)^{1/4} \left(\frac{1/2}{1} \right)^{1/2} \left(\frac{300}{1/4 + 1/2} \right)^{3/4 + 1/2} = 12.80$$

Costs / Expenditure:

$$\text{Costs } e(p, u) = p_1^{\frac{\alpha}{\alpha + \beta}} p_2^{\frac{\beta}{\alpha + \beta}} u^{\frac{1}{\alpha + \beta}} \left[\left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha + \beta}} + \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha + \beta}} \right]$$

Costs under initial prices, p^* and maximal utility level $u^* \equiv v(p^*, w)$

$$e(p^*, u^*) = 1^{\frac{1/4}{1/4 + 1/2}} 1^{\frac{1/2}{1/4 + 1/2}} (15.81)^{\frac{1}{1/4 + 1/2}} \left[\left(\frac{1/4}{1/2} \right)^{\frac{1/2}{1/4 + 1/2}} + \left(\frac{1/2}{1/4} \right)^{\frac{1/4}{1/4 + 1/2}} \right]$$

$$= 1^{1/3} 1^{2/3} (15.81)^{4/3} \left[\left(\frac{1}{2} \right)^{2/3} + (2)^{1/3} \right]$$

$$= 299.96 \approx 300$$

Costs under initial prices, p^0 and maximal utility level, $u^0 \equiv v(p^0, w)$,

$$e(p^0, u^0) = 4^{1/3} 4^{2/3} (18.80)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 377.895$$

Costs under initial prices, p^1 and maximal utility level, $u^0 \equiv v(p^1, w)$,

$$e(p^1, u^0) = (2)^{1/3} 4^{2/3} (15.81)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 238.08$$

Costs under new prices p^1 and maximal utility level, $u^1 \equiv v(p^1, w)$

$$e(p^1, u^1) = 2^{1/3} 4^{2/3} (18.80)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 299.94 \approx 300$$

iii) Equivalent variation: (EV)

The amount of money that the consumer would be willing to give up before the price increase to be as well off as after the price increase.

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0) = 377.895 - 300 = 77.9 \approx 78$$

Compensated variation (CV):

The amount of money that a consumer would need after a price increase to be as well off as before the price increase.

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0) = 300 - 238.08 = 61.92 \approx 62$$

Consumer surplus: (CS)

Price of commodity 1 decreased by 50%, from $p^0=4$ to $p^1=2$.

The change in consumer surplus is:

$$\begin{aligned}
 CS &= \int_2^1 x_1(p, w) dp_1 \quad | \quad x_1(p, w) = \frac{1}{3} \frac{m}{p_1} \rightarrow \text{Walrasian demand} \\
 &= \int_2^1 \left(\frac{1}{3} \frac{m}{p_1} \right) dp_1 \\
 &= \int_2^1 \left(\frac{1}{3} \times \frac{300}{p_1} \right) dp_1 \\
 &= 100 \int_2^1 \frac{1}{p_1} dp_1 \\
 &= 60.21
 \end{aligned}$$

Relationship between EV , CV and CS :

$$EV > CS > CV$$

Nature of commodities:

As $EV > CS > CV$, for the decrease in price of commodity 1, the commodities are normal goods.